

COMPRITOL® 888 ATO VS PRECIROL® ATO 5 IN HOT MELT EXTRUSION: INVESTIGATION OF OPERATING WINDOW LIMITS

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INTRODUCTION

Solid lipid excipients are interesting materials for hot melt extrusion for a number of reasons. They are functional excipients allowing the modulation of drug dissolution properties; they can be extruded at relatively low temperatures (below 70°C); and they recrystallize rapidly on cooling, resulting in hard extrudates. Excipient functionality depends on chemical composition. For example, Compritol® 888 ATO and Precirol® ATO 5 are fatty esters with long acid chain length and low hydrophilic-lipophilic balance. They are water insoluble and can be used to form sustained release matrices. Our study was designed to evaluate the operating window of both excipients and to compare their processability in hot melt extrusion.

EXPERIMENTAL METHODS

Compritol® 888 ATO (glyceryl behenate NF) and Precirol® ATO 5 (glyceryl distearate NF) were extruded using a Pharma 11 HME and the standard screw design by Thermo Fisher (Figure 1). The barrel temperature profile towards a 1mm die was fixed to: 60-63-68-70-65-70-70 and 45-45-50-53-50-53-54-54 for Compritol® 888 and Precirol® respectively. Feed rate and screw speed were increased step wise until the process limit was reached. Good processability was concluded if smooth extrudates were obtained continuously. In the second part of the study niacin was extruded with each excipient. The niacin: lipid ratio 60:40 (% w/w) was exemplarily depicted to evaluate the impact of drug incorporation on the investigated process limits for each lipid. Finally, we determined the maximum possible niacin load and the effect on drug release after compression into tablets (900ml, 0.1N HCl, 37°C).

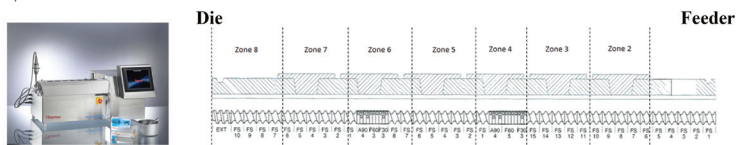


Figure 1: Thermo Scientific Pharma 11 HME (Thermo Fisher Scientific) and the adapted screw design.

RESULTS AND DISCUSSION

Figure 2a indicates that the operating window (under the investigated screw design and temperature profile) is substantially smaller for Precirol® than for Compritol® 888. It was noted that Precirol® needs slightly more time to crystallize after extrusion from the die, resulting in a reduced working space. Figure 2b shows that when 60% niacin is extruded in Compritol® 888 the working space does not change significantly. In contrast, the operating window of Precirol®: niacin broadens notably as illustrated in Figure 2c. It becomes even larger than for Compritol® 888: niacin (Figure 2d). The addition of the high drug percentage results in harder extrudates with a wider processing range.

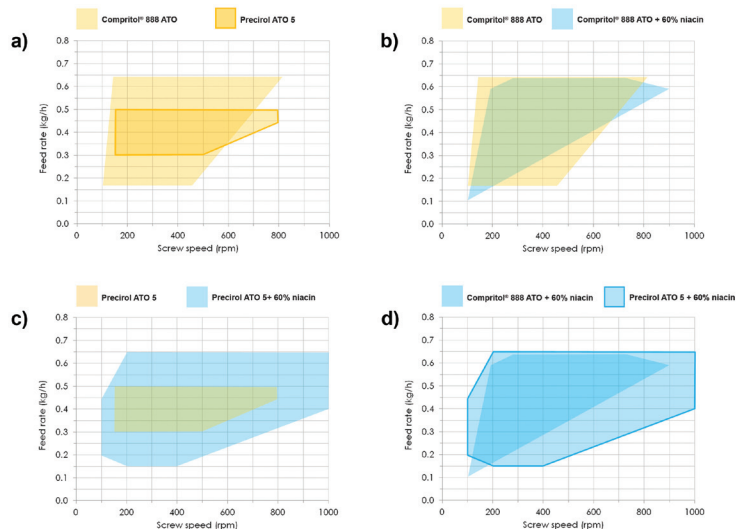


Figure 2: Potential operating window of: a) Compritol® 888 vs Precirol®, b) Compritol® 888: 60% niacin, c) Precirol®: 60% niacin and d) Compritol® 888: 60% niacin vs Precirol®: 60% niacin under the described process conditions.

The highest possible niacin load for Precirol® was 70% by weight. Above this concentration the production of extrudates became discontinuous. In contrast, a 75% niacin load (by weight) was achieved with Compritol® 888. Figure 3 shows the drug release kinetics from tablets made with the extrudates. Even with a high drug load (75%) Compritol® 888 provides extended release over 12 hours (Figure 3a). Surprisingly however drug release from Precirol® matrices was much slower compared with Compritol® 888 (Figure 3b), which is not the case with conventional production methods^{1,2}. Similar tendencies have been reported in the literature when hot preparation methods were applied instead of direct compression or wet granulation^{3,4}.

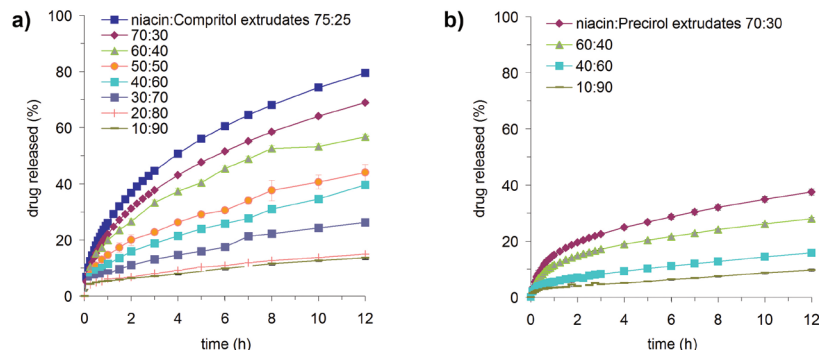


Figure 3: Niacin release from tablets obtained by the compression of: a) Compritol® 888: niacin extrudates and b) Precirol®: niacin extrudates.

The optimal extrusion parameters enabling the production of high dose extrudates continuously are summarized in the following Figure 4.

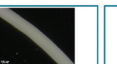
Ratio	Niacin:Compritol® 888 70:30			Niacin:Precirol® 70:30		
Feed rate	100 g/h	300 g/h	600 g/h	100 g/h	300 g/h	600 g/h
Screw speed	150 rpm	300 rpm	400 rpm	170 rpm	350 rpm	480 rpm
Temperature	60-63-68-73-68-73-71-70°C	60-63-68-73-68-73-72-70°C	60-63-68-73-68-72-71-71°C	45-45-50-53-50-56-55-53°C	45-45-50-53-50-55-54°C	45-45-50-53-50-55-55°C
Extrudate aspect						

Figure 4: Processing guideline facilitating Compritol® 888 extrusion at different niacin concentrations (temperature from feed to die).

CONCLUSION

This study was designed to evaluate the operating window for the melt extrusion of lipid excipients with active pharmaceutical ingredient. The assessment of process flexibility and the identification of robust process parameters enable faster, more error-free drug production. The functional excipients used enable the production of sustained release dosage forms. Although the processing flexibility with Precirol® alone was limited, the larger melting and crystallization range proved to be advantageous when API was added. Although their chemical composition is different, Compritol® 888 and Precirol® behave rather similarly under the investigated process conditions. The results from this study facilitate the development of processing guidelines with these excipients in hot melt extrusion.

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